

given price point. They typically run at 7200 RPM and have multiple platters onto which data is stored. Standard capacity ranges from 500 GB to 4 TB.

2.5"

Another all too common form is the 2.5" drive which is standard in laptops and portable storage drives. They are low power drives with a single platter and don't boast of much storage space. Recently, these drives have breached the 1 TB capacity limit.



1.0"

If you've wondered how miniature devices like PMPs boast of over a 100 GBs of storage space then this is the answer. The microdrive used to be limited to roughly 10-20 GBs but over the years we have seen capacities like 160 GB iPods which had microdrives in them.

Capacity

The early hard drive just had the one platter and two heads that simultaneously accessed the top and bottom sides of the platter. With improvements in technology the data that could be squeezed onto the platter increased tremendously. This is termed as density. Once the upper limits of density was reached the only step forward was to have multiple platters. This further boosted storage space to the extent that we now have 3.5" drives that can store 4 TB of data.

Speed

The speed at which data is accessed comes down to the RPM at which the disc spins at. The following table lists the range of common RPM values.

Form factor	RPM range
3.5" SATA	5400 - 10000 RPM
3.5" SCSI or SAS	10000 - 15000 RPM
3.5" PATA	7200 - 10000 RPM
2.5" SATA or PATA	5400 RPM